

ece4-recipe

Effortless EC-Earth4 Configuration Management

A standalone toolkit for reproducible climate experiment setups.

The Problem: Configuration Hell

- **Complexity:** EC-Earth4 YAMLs are powerful but complex.
- **Fragmentation:** Duplicate configs across experiments.
- **Dependency:** Hard to test or generate configs outside of a heavy workflow manager.
- **Traceability:** Hard to know "exactly" what went into a configuration.

The Solution: **ece4-recipe**

A Modular, Standalone Tool

- **Decoupled:** Works anywhere (Local, MN5, ECMWF).
- **Stitched:** Uses a "Merge Model" to layer configurations.
- **Versioned:** Automatically syncs with specific EC-Earth4 source tags.
- **Reproducible:** Save your modifications as a minimal "Recipe".

How it Works: The Merge Model

Configurations are layered like a cake:

1. **[BASE]** experiment-config-example.yml (from ECE4 repo)
2. **[PLATFORM]** Slurm/HPC settings (MN5, HPC2020, etc.)
3. **[RECIPE]** Your scientific setup (gcm-sr.yml, omip-hr.yml)
4. **[OVERRIDE]** Your specific experiment tweaks.

Demo: Instant Configuration

No dummy files needed. Just one command:

```
./ece4-recipe.sh generate --dry-run \  
  --repo-owner ec-earth \  
  --repo-branch v4.1.6 \  
  --platform bsc-marenostrum5 \  
  --sim-procs 1120 \  
  --kind auto \  
  --launcher slurm-wrapper-taskset \  
  --recipe gcm-sr.yml
```

Result: A fully verified, 10-node Slurm configuration ready to go.

Key Features for Researchers

- **Quick Validation:** Catch syntax or schema errors instantly.
- **Recipe Sharing:** Save your custom setup as a tiny YAML fragment.
- **HPC Ready:** Pre-configured for BSC and ECMWF.
- **Bash Completion:** High developer ergonomics.

Sharing Science: Save your Recipe

Modified your experiment and found a perfect setup?

```
# Saves only your changes relative to the base  
./ece4-recipe.sh save my-breakthrough-setup.yml
```

Result: A clean, shareable recipe that anyone can use to reproduce your experiment exactly.

Get Started Today

Github: <https://github.com/vlap/ece4-recipe>

```
git clone https://github.com/vlap/ece4-recipe.git
./ece4-recipe.sh init-user
./ece4-recipe.sh list
```

Questions?

vladimir.lapin@bsc.es